Lesson notes · for you

Operations research: unit check

Optimization · Level 4 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Linear programming and the simplex method
- Integer programs and branch and bound
- Constraint programming: propagation and search
- Heuristics and trade-offs

Activity: Activity: Pick the tool

1. Sort (6 min): On cards: a canteen menu, buses for a trip, an exam timetable, a 300-stop delivery round, a farm plan, a hospital roster, a sudoku, choosing warehouse sites, a study plan, a sports league. Put each under LP, MIP, CP-SAT or heuristic, with one reason.
2. Model (8 min): Pick one card and write its variables (with units), constraints and objective in one paragraph.
3. Report (6 min): List what a decision-maker should see besides the plan: binding limits and shadow prices, the gap, the assumptions, and the trade-offs and fairness limits.

Many problems fit more than one tool; the reason matters more than the label.

Discussion

- When is a relaxation's answer useful, even though it can't be used as it is?
- Why is INFEASIBLE good news for a modeller?
- What does a responsible modeller owe the people affected by a plan?

Check yourself

- Name one problem from your own life, the tool you'd use, and the one number you'd report besides the answer.

Try it this week

Take one decision you care about (a society budget, a timetable, a delivery round) and decide which tool fits it: LP, MIP, CP-SAT or a heuristic. Write the model's variables, constraints and objective in one paragraph, and say what you would report besides the answer: binding limits, the gap, the trade-offs.

Worksheet

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Name _____ Class _____ Date _____

1 **Why does a linear program with an optimum always have one at a corner of the feasible region?**

- ◯ The region is convex and the objective is linear
- ◯ Solvers only look at corners
- ◯ Corners always have whole-number values
- ◯ The constraints are all equalities

2 **For a minimisation MIP, the LP relaxation gives an upper bound on the optimal cost.**

- ◯ True
- ◯ False

3 **Match each method to the problem it suits best.**

Simplex (GLOP)	• • • •	A 300-stop delivery route by morning
Branch and bound (SCIP)	• • • •	Exam slots with all-different rules
CP-SAT	• • • •	Whole buses with a booking fee
2-opt local search	• • • •	The diet problem with fractional portions

4 **What does this program print?**

```
found, bound = 1400, 1173.33
gap = (found - bound) / found
print(round(100 * gap))
```

- ◯ 16
- ◯ 19
- ◯ 84

5 **Which statements about shadow prices are true? Choose all that apply.**

- ☐ A binding constraint can have a non-zero shadow price
- ☐ A constraint with slack has a shadow price of zero
- ☐ A shadow price holds for any change, however large
- ☐ In the garden, 8 × 25 + 18 × 5 + 14 × 0 equals the best profit, 290
- ☐ Shadow prices tell you which variables must be whole numbers